

VR Shooter (First Encounters)

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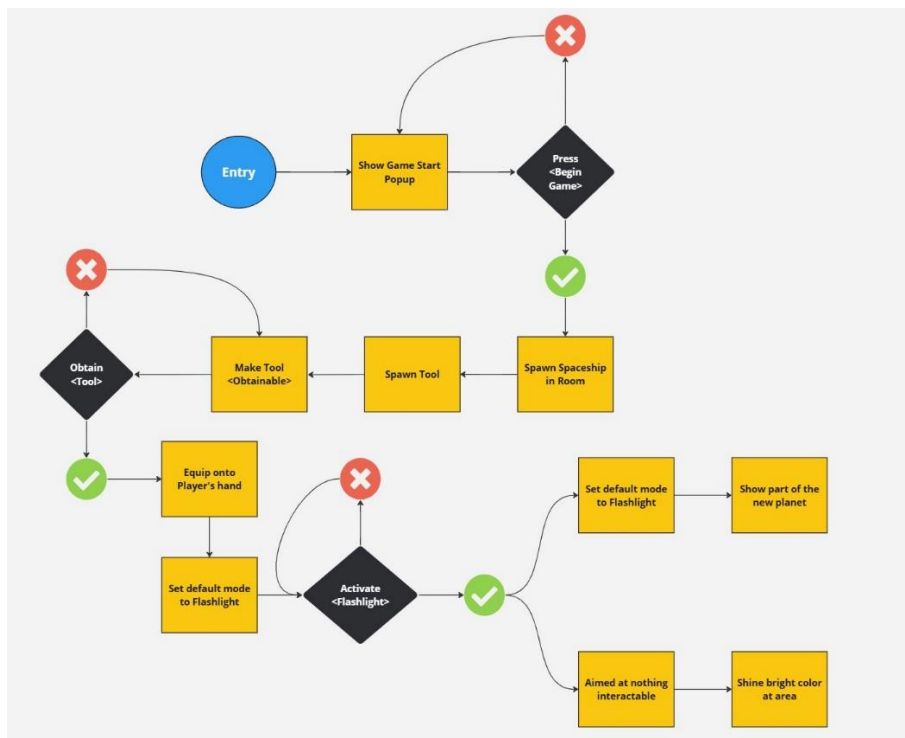
Introduction

First Encounters is a first-person shooter game for the Meta Quest 3 that comes built in for each device. This game takes place in a new planet where “puffians” swarm your surrounding, you aim to shoot/catch as many space “puffians” as you can (Meta).

I will be discussing the user flows for 3 tasks, my experience, the benefit(s), about the overall user interface / user experience (UI/UX), other thoughts and my conclusive thoughts.

User Flow

1: Using the Flashlight



To Activate the flashlight, the user must press the trigger. To get to this step, we follow this series of steps.

Users Perspective:

1. Open the game First Encounters
2. Press the Begin Game button
3. Grab the Tool
4. Press the trigger to activate the Flashlight (LOOP)
 - a. IFF the flashlight is pointed at nothing, just light shines normal
 - b. IFF the flashlight is pointed at interactable, show part of the strange world

System Perspective:

1. Load the < Begin Game > popup
2. Wait for user interaction for player to press the <Play> button
 - a. IFF they do not press < Play >, keep waiting
 - b. IFF they press < Play > GOTO step 3
3. Load the Spaceship into the room
4. Spawn the tool above the spaceship
5. Set the tool to be obtainable & interactable
6. Wait for user interaction for player interacting with the < tool >
 - a. IFF the user doesn't obtain it, keep waiting for them to interact with it
 - b. IFF the user interacts with it, GOTO step 7
7. Equip it onto the player's hand
8. Set the default mode to flashlight
9. Wait for user interaction for user to press the flashlight trigger

a. IFF they don't press the trigger to shine the light, keep waiting

b. IFF they press the trigger, GOTO step 10

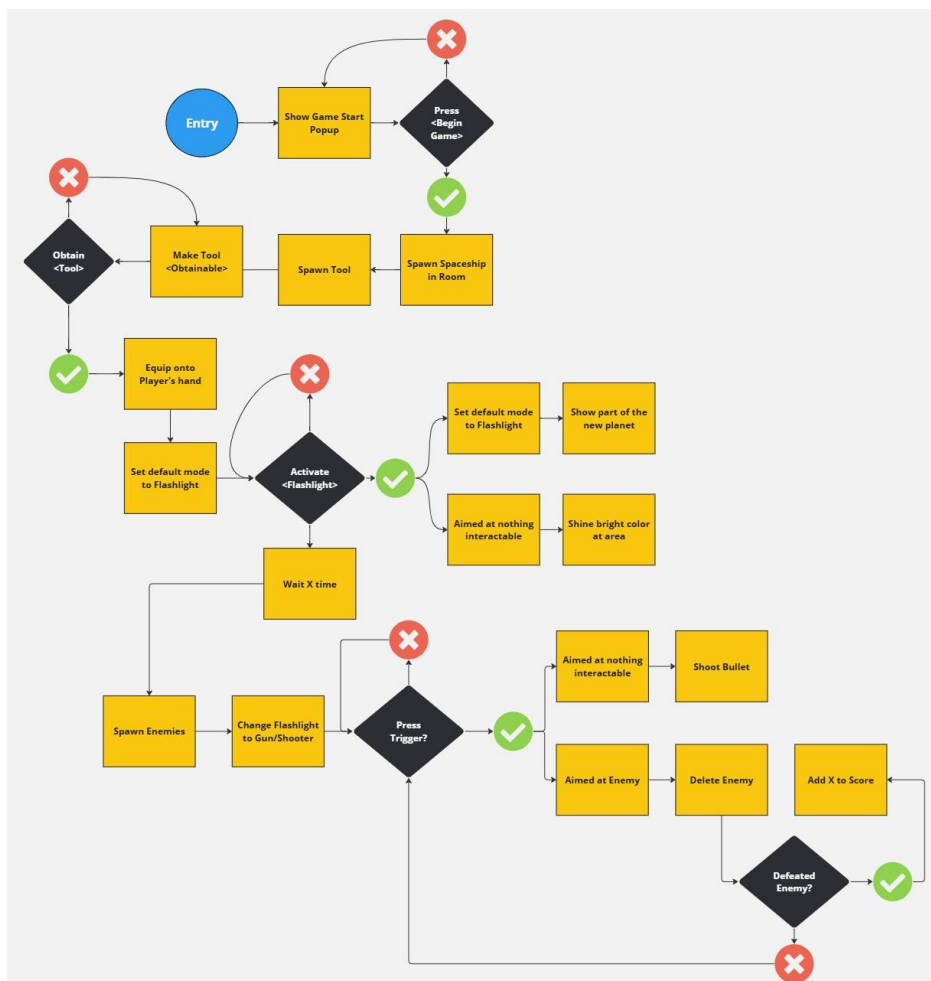
10. Follow 1 of the two scenarios

a. IFF the light is shined at nothing interactable, shine the nice colour

b. IFF the light is shined at something interactable, show a part of new world

11. GOTO Step 9

2: Using the Shooter/Gun



To Activate the Gun/Shooter, the user must press the trigger. To get to this step, we follow this series of steps.

Users Perspective:

1. Open the game First Encounters
2. Press the Begin Game button
3. Grab the Tool
4. Press the trigger to activate the Flashlight
 - a. IFF the flashlight is pointed at nothing, just light shines normal
 - b. IFF the flashlight is pointed at interactable, show part of the strange world
 - c. Simultaneously, start Step 5+
5. After X amount of time game will change flashlight to gun/shooter
6. Press the trigger to activate the gun/shooter (LOOP)
 - a. IFF the gun is pointed at an enemy, shoot the projectile then damage the enemy
 - i. IFF enemy is defeated, add X Score
 - ii. IFF enemy is NOT defeated, wait for user input
 - b. IFF the gun is pointed at nothing, shoot the projectile

System Perspective:

1. Load the < Play Game > popup
2. Wait for user interaction for user to press the <Play> button
 - a. IFF they do not press < Play >, keep waiting

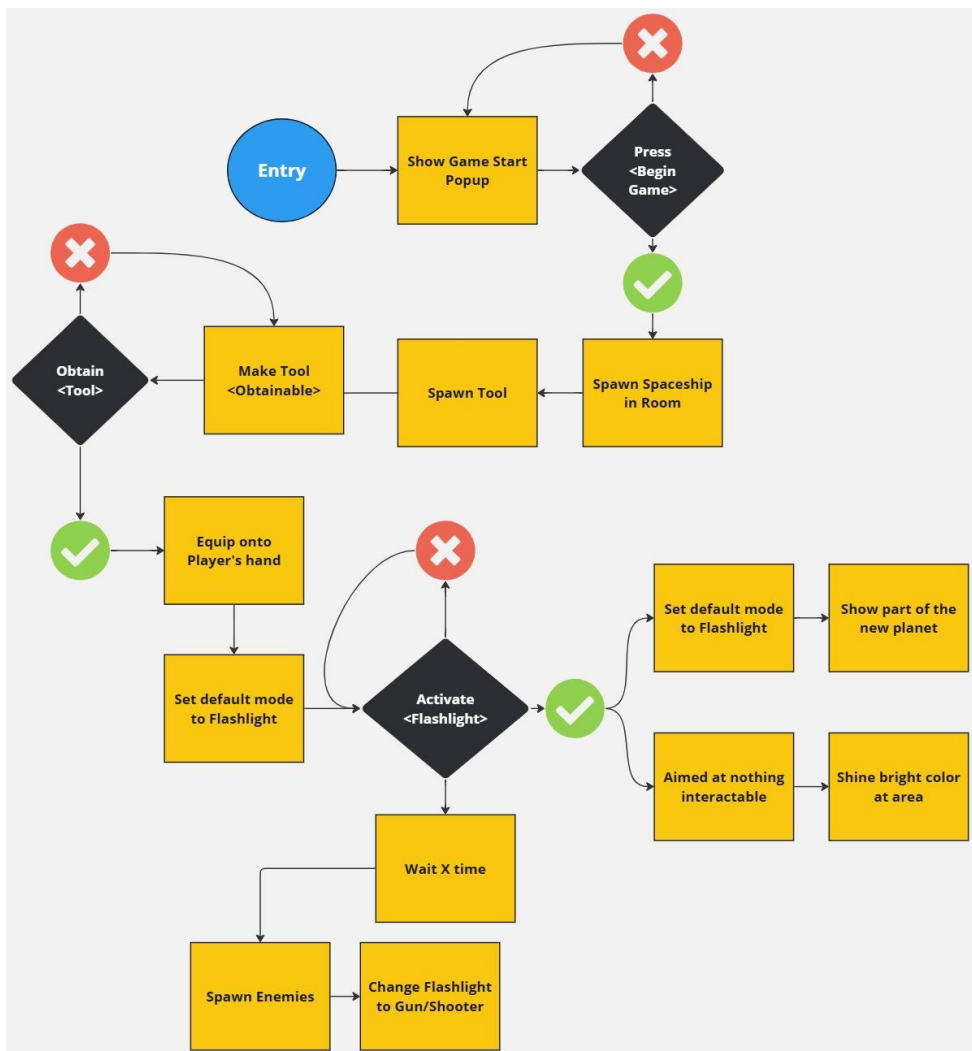
- b. IFF they press < Play > GOTO step 3
3. Load the Spaceship into the room
4. Spawn the tool above the spaceship
5. Set the tool to be obtainable & interactable
6. Wait for user interaction for player interacting with the < tool >
 - a. IFF the user doesn't obtain it, keep waiting for them to interact with it
 - b. IFF the user interacts with it, GOTO step 7
7. Equip it onto the player's hand
8. Set the default mode to flashlight
9. Wait for user interaction for player pressing flashlight trigger
 - a. IFF they don't press the trigger to shine the light, keep waiting
 - b. IFF they press the trigger, GOTO step 10
 - c. Simultaneously, start Step 11
10. Follow 1 of the two scenarios
 - a. IFF the light is shined at nothing interactable, shine the nice colour
 - b. IFF the light is shined at something interactable, show a part of new world
11. Wait X time, then continue flow
12. Spawn enemies
13. Change player's tool from flashlight to gun/shooter
14. Wait for user interaction for player pressing the gun trigger (LOOP)
 - a. Player Press the trigger on nothing
 - i. Shoot projectile

b. Player press trigger at enemy

- i. Shoot projectile
- ii. Damage Enemy
- iii. Check if enemy is defeated
 1. IFF yes: GOTO step 15
 2. IFF no: GOTO Step 14

15. Add X to Score and GOTO Step 14

3: Getting the Gun



To get the Gun/Shooter, the user must get the flashlight then wait X amount of time. To get to this step, we follow this series of steps.

Users Perspective:

7. Open the game First Encounters
8. Press the Begin Game button
9. Grab the Tool
10. Press the trigger to activate the Flashlight
 - a. IFF the flashlight is pointed at nothing, just light shines normal
 - b. IFF the flashlight is pointed at interactable, show part of the strange world
 - c. Simultaneously, start Step 5+
11. After X amount of time game will change flashlight to gun/shooter

System Perspective:

16. Load the < Play Game > popup
17. Wait for user interaction for user to press the <Play> button
 - a. IFF they do not press < Play >, keep waiting
 - b. IFF they press < Play > GOTO step 3
18. Load the Spaceship into the room
19. Spawn the tool above the spaceship
20. Set the tool to be obtainable & interactable
21. Wait for user interaction for player interacting with the < tool >
 - a. IFF the user doesn't obtain it, keep waiting for them to interact with it

- b. IFF the user interacts with it, GOTO step 7
- 22. Equip it onto the player's hand
- 23. Set the default mode to flashlight
- 24. Wait for user interaction for player pressing flashlight trigger
 - a. IFF they don't press the trigger to shine the light, keep waiting
 - b. IFF they press the trigger, GOTO step 10
 - c. Simultaneously, start Step 11
- 25. Follow 1 of the two scenarios
 - a. IFF the light is shined at nothing interactable, shine the nice colour
 - b. IFF the light is shined at something interactable, show a part of new world
- 26. Wait X time, then continue flow
- 27. Spawn enemies
- 28. Change player's tool from flashlight to gun/shooter

VR Experiences

My feeling using VR isn't a very pleasant one, as even though I was excited and it is very easy to use, it is very physically exhausting. Whenever I used VR, I always got a headache and so the experience wasn't fully enjoyable due to that. Additionally, since I wear glasses, it makes for a much more uncomfortable experience as it doesn't feel comfortable to wear both my glasses and VR devices. However, those are mostly outside factors, the VR itself, is fun, unique and easy to use. Personally, I would rather

not use it again, since as I've mentioned, headaches, uncomfortable experiences due to glasses and just I don't enjoy knowing a screen is extremely close to my eyes and face. This isn't to say the experience itself wasn't fun, I just don't see the enjoyment from VR to be enough to comprise for its disadvantages, especially since I don't care too much for it.

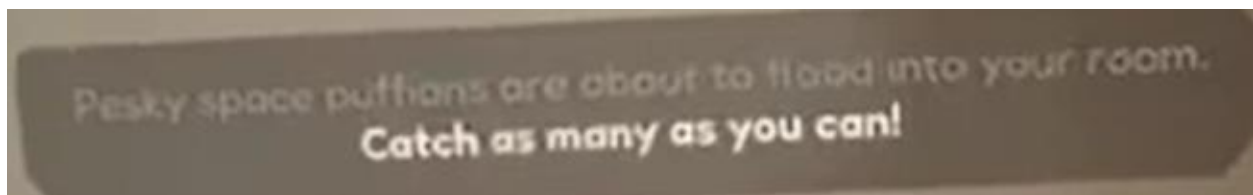
Benefit:

The major benefit of VR is the immersive experience it brings. When compared to other physical or even digital products, they are inherently less immersive since VR's whole gimmick is in the name, VR, Virtual Reality, it focuses entirely on transforming your average experience into a much of immersive one by allowing you to interact with consumables and changing your entire environment around you to fit the need. For instance, if one were to play a roller coaster game on desktop, you are simply looking at your monitor whilst the animation plays, but your surrounding is still your own setup, you see that around you is a keyboard, chair, desk, etc However, with VR, those same surroundings would be the inside of a roller coaster, since VR aims to give a more immersive experience, they change the surroundings of your entire environment itself and lets you visualize the entire experience of a roller coaster without seeing anything that would break the immersive experience. Another example can be in an educational setting, there are VR games that simulate the experience of for instance, a registered nurse. In games like these, you can practice day to day activities and even things like CPR all in a virtual world without running the risk of causing any real harm (Meta). Statistically, through the addition of this VR Nurse application that was used by Purdue-

Global, there was a “10-15% increase [in the] pass rate on the national exam to become a registered nurse” (Meta). This shows that the major benefit of immersive capabilities not only increases the enjoyment of consumers but also can provide educational benefits by allowing for them to have a more immersive and real experience.

UI Design:

For my application, First Encounters, it follows these usability heuristic, visibility, consistency and standards, user control and freedom and aesthetic and minimalist design. Firstly, for visibility of the system, the game has popup text to explain to the user the current situation. For instance,



The image above displays at the start of the game, it tells users about the current state of the game. It states that the space puffians are '**about**' to flood into their room and to “catch as many you can”, it keeps the user informed about what will happen and does this at the start to keep the user on track. Another example is this:



The game shows the score to keep the user updated and knowing about their current progress throughout the game. Secondly, consistency and standards. The game follows a standard convention of displaying score and keeps the colours all visible. Also, it keeps the conventions for things like combos associated to the letter 'x' and having the word combo, as you can see from the image below.



Thirdly, I wanted to discuss the user control and freedom, the user is allowed for act as they please, they can move around and do pretty much any action they want as all the error actions have already been handled and the user will not run into any of them. Finally, I wanted to talk about aesthetic and design. The game has a very simplistic design of only a couple of colors and keeps everything relatively minimal, even the enemies have a simplistic design and so does the gun as well. They all have nothing too extraordinary and doesn't take away from the user's attention at all.

Now, while it does have these good qualities, there is 2 heuristics I would suggest them to work on, them being, visibility and flexibility. I know I said visibility was good, but it doesn't mean there isn't room for improvement, mainly because during the middle of the game, there is lack of indication of what to do. There should be a mission bar that tells the user what to do, for instance, when the gun appears, the user isn't told anything, they would have to intuitively understand it rather than being told to. Then comes flexibility, the application doesn't really have any way for an advanced user to do anything faster than a beginner and as a beginner it's not too clear for the user. I would suggest just to add something near the top right or bottom left that shows the current 'mission' or progress.

Other thoughts:

I don't have much to say here expect, even though the VR gives the options to expand to adjust for your glasses, it still isn't that comfortable especially when compared to not using glasses.

Conclusion:

Overall, First Encounters is a fantastic game as it allows for a pleasant use of VR making use of a lot of capabilities to make an interactive, immersive and fun game. It follows the visibility, consistency and standards, user control and freedom, aesthetic and minimalist design principals but doesn't follow the visibility of the system heuristic too well.

References

Meta. (n.d.). *First encounters on Meta Quest*. <https://www.meta.com/en-gb/experiences/first-encounters/6236169136472090/>

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